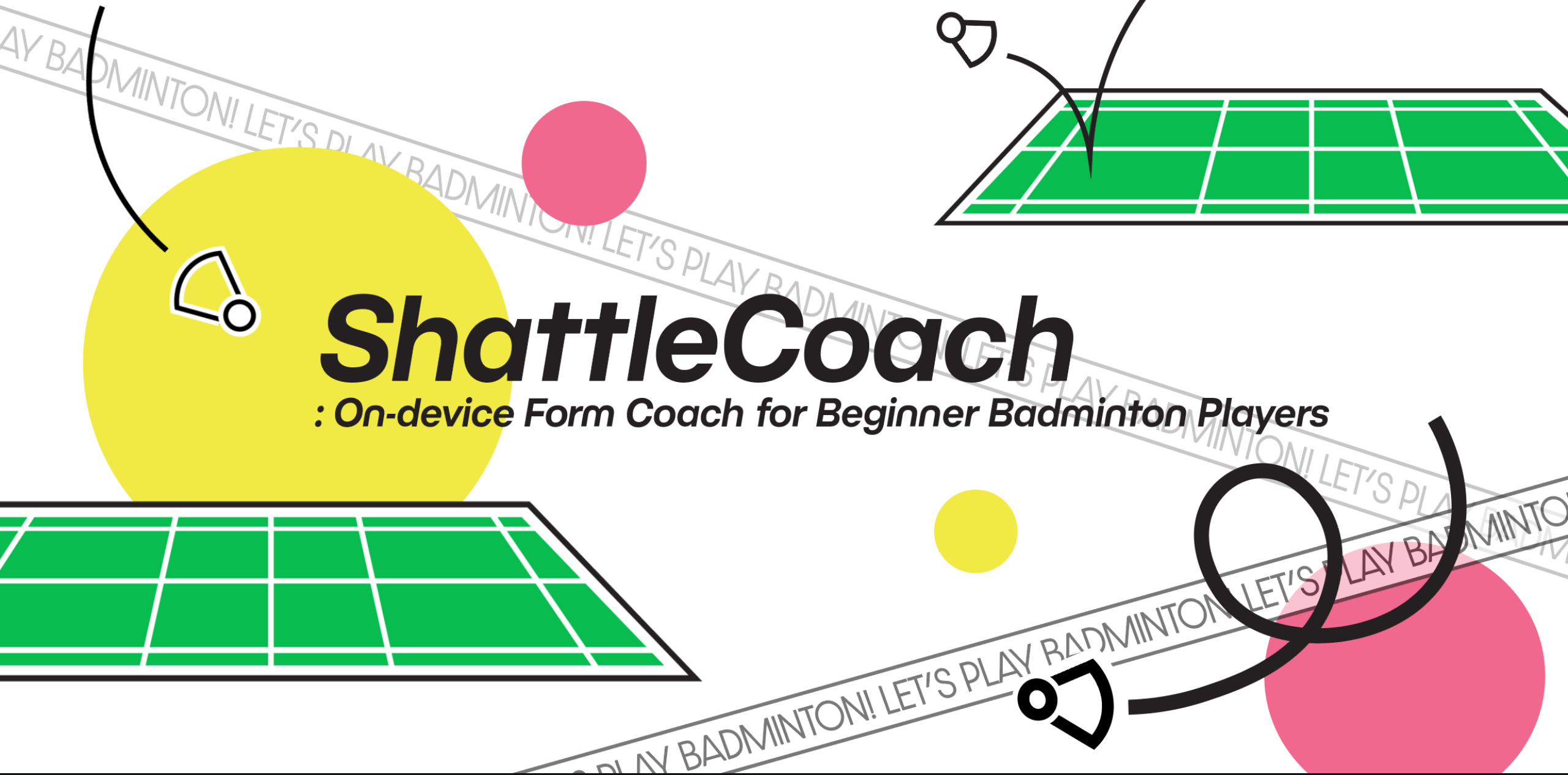
Eunwoo Choi, Donghwi Moon



ShattleCoach

: On-device Form Coach for Beginner Badminton Players

GOAL & USAGE SCENARIO

01

Coaching that doesn't scale — solved on the phone

The problem

University clubs absorb dozens of beginners each semester, but a few seniors can't watch every swing. Bad habits — frypan grip, dropped elbow, late impact — set in within a month.

Our goal

From a 5-second swing clip, return per-criterion form feedback for three foundational strokes — **all inference on-device, no video leaves the phone.**

Built for real use at Shattlecock (SNU CSE badminton club): form coaching + everything club operations needs.

Usage scenario

1. **Mount & pick**
   Tripod ~3 m from the baseline, open the app, choose a stroke.
2. **Record 5 s**
   Live pose-skeleton overlay confirms a good side-on framing.
3. **On-device analysis**
   MoveNet + scorer run locally (~24 s on Galaxy S21).
4. **Read the scorecard**
   Stars, impact-frame annotation, and a short Korean coaching note.

KEY IDEA

A three-stage on-device pipeline

02

1 · Pose extraction

MoveNet Lightning (TFLite INT8, ~2.9 MB) → 17 COCO keypoints per frame, drawn live on the preview.

→

2 · Form scoring

Rule-based geometry on 4 criteria (impact point · elbow · waist rotation · knee) + BST percentile lookup for speed & footwork. Axes adapt per stroke.

→

3 · Visualization

Per-axis 1–5★, an annotated impact frame, “what you did well,” and a deterministic Korean coaching note (optional LLM polish).

Three strokes, each with its own scoring axes

High clear **Short serve** **Forehand drive**

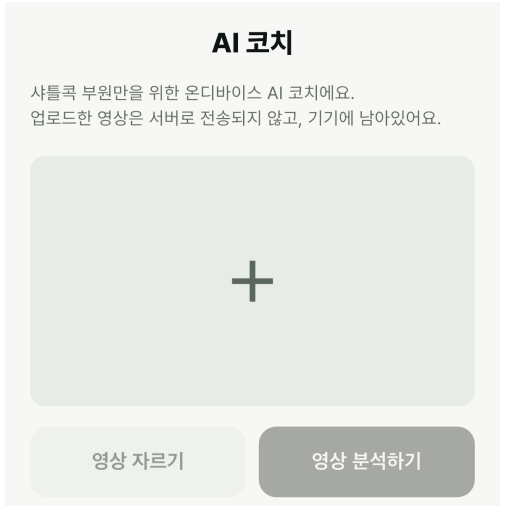
Stroke-conditional axes (e.g. footwork is dropped for short serve) keep each score meaningful.

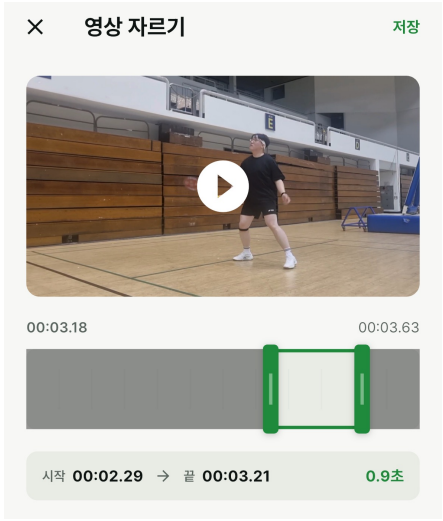
Why pose input?

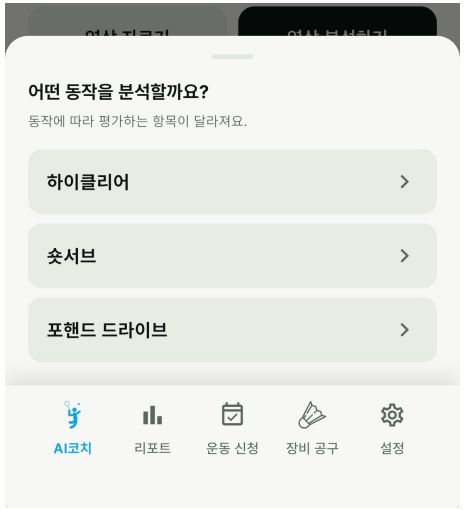
5 s of HD video → ~5 KB of keypoints. Privacy by design (nothing stored or sent) and small enough for real-time on-device inference.

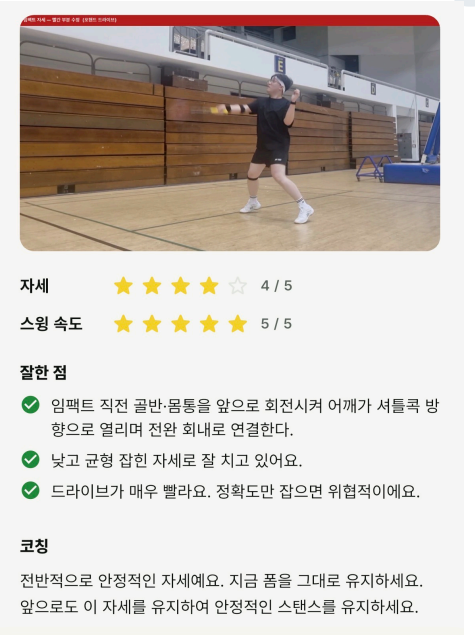
APP FUNCTIONALITY · 1

AI Coach — the core loop









Select — gallery clip, 5 s+ → **Trim** — drag the swing window → **Choose stroke** — 3 options → **Analyze** — on-device

The scorecard

Per-axis star ratings

Posture (4-criteria geometry), swing speed & footwork (BST percentiles).

Impact-frame annotation

The impact moment with circles, arrows and a Korean cue overlaid.

What you did well

1–3 short positive lines with check icons — encouragement first.

Korean coaching note

Deterministic rulebook sentence; optional Groq llama-3.1-8b polish, guardrailed (no new advice).

DEMO

See it in action

07

YouTube Short demo

youtube.com/shorts/_IbC8qLcBfQ

The full user journey

1. **Onboarding**
   Google login → intro → profile (name, handedness)
2. **AI Coach**
   Pick clip → trim → choose stroke → on-device scorecard
3. **Reports**
   Monthly heatmap of progress; tap a day for records
4. **Sessions**
   Browse the week, RSVP / cancel practice slots
5. **Equipment**
   Group-buys & member sharing board

Captured on a iPhone 15 · scan or open the link for the 30-second walkthrough.

APP FUNCTIONALITY · 2

Not just a coach — a club platform

ShattleCoach already runs Shattlecock's day-to-day. Five tabs, one app, Google login over Supabase.

AI Coach

Swing video → on-device form scorecard.

Reports

Monthly heatmap calendar of practice & progress; tap a day for records.

Sessions

Weekly schedule with RSVP / cancel, capacity, attendee list (Supabase).

Equipment

Community board: official group-buys (exec-only) and member sharing.

Settings

Profile, executive-code registration, OAuth, licenses.

Auth & roles

Google OAuth + guest (grader) login; row-level security & exec roles.

Beyond the assignment: *a real product we're polishing to actually manage the club next semester.*

Surviving the real world

Our ST-GCN stroke classifier scored 89.6% on broadcast data but collapsed to 28.6% on real phone clips — a domain gap (the model leaned on a confidence=1 crutch absent in noisy MoveNet keypoints).

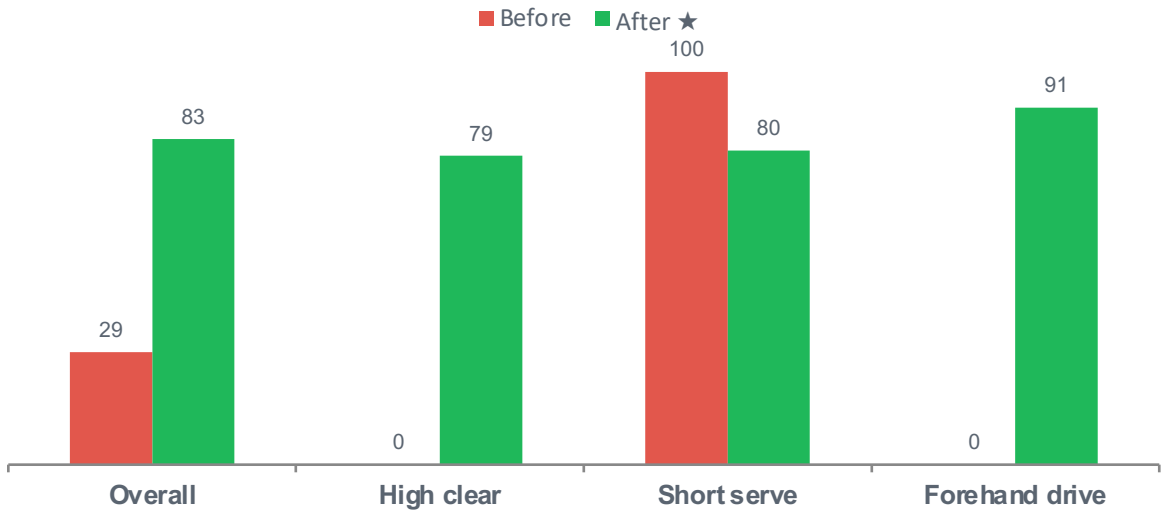
28.6% → **82.9%**

before (real phone) — **after retraining**

Honest trade-off: broadcast accuracy fell 89.6% → 49.4% — the model became a "phone expert," which is all the app sees (test n=35, small).

How: domain-adaptation retraining

- Randomize keypoint confidence + random joint-drop → kill the conf=1 crutch
- Augment to mimic MoveNet noise (flip · time-warp · jitter · drop)
- Oversample real phone clips; boost the weak class (1.8× loss)



Optimizing for the phone: size & speed

Quantization — the real work was making INT8 run on mobile

INT8 scheme	Size	Accuracy	On mobile onnxruntime
Naive INT8 (poor calibration)	327 KB	−20.89 pp	rejected — too costly
Dynamic INT8 (weight-only)	308 KB	89.7% (desktop)	**ConvInteger unsupported → crash**
QDQ static INT8 ★	**327 KB**	**−2.9 pp (80.0%)**	**QuantizeLinear → runs ✓**

3.2× smaller (1,061 → 327 KB) at only −2.9 pp once calibration was fixed and Conv wrapped in Q/DQ instead of the unsupported ConvInteger.

Latency — 3-isolate parallel inference

51 s → 24 s

End-to-end on Galaxy S21, 1 → 3 Dart isolates (2.1× faster) — what makes on-device analysis practical.

Production decision: the shipped scorer is rule-based (geometry + BST percentiles) for out-of-distribution robustness; the optimized neural classifier is retained for a "stroke-confirmation" feature and future release.
Optimization scoring focuses on size-reduction rate and the marginal −2.9 pp drop.

Five weeks, two people

Project timeline (May 5 – Jun 1)

W1 Rubric & repo; pilot labeling; Flutter scaffold + pose overlay

W2 Filming session 1; FP32 training; recording flow + scorecard shell

W3 Filming session 2; baseline CV; on-device inference wired

W4 Retraining & quantization study; latency / size benchmarks

W5 Final report & figures; UI polish; APK build; demo video

Contribution

Both — self-collected, double-labeled swing dataset (filming + annotation).

Eunwoo Choi — App & integration & UI Design

Entire Flutter app: UI, routing, auth (Google/Supabase), Reports / Sessions / Equipment / Settings, native frame-extractor channels, wiring the ML pipeline into the app end-to-end.

Donghwi Moon — On-device ML

Model training & evaluation, domain-adaptation retraining, quantization & ONNX experiments, and the scoring + coaching logic.

Stack: Flutter 3.44 · Riverpod · go_router · Supabase (Postgres/Auth/RLS) · MoveNet INT8 (TFLite) · rule-based scoring + BST percentiles